1. $$\gcd(a,b) = 1, \qquad a \mid bc \quad\Longrightarrow\quad a \mid c$$

   Prove it from Bézout's identity rather than prime factorization.

2. Prove that there are infinitely many primes

   $$p \equiv 3 \mod 4$$

   A Euclid-style argument works, but simply taking the product of known $3 \bmod 4$ primes and adding 1 does not give the useful construction directly. Try to determine what number to construct and why at least one of its prime divisors must be $3 \bmod 4$.